

General Instructions to the candidate

- Please write your answers legibly and neatly in the answer booklet provided only.
- Symbols, constants, terminology used have their usual meaning unless specified specially.
- There are a total of **Four** questions, and all are compulsory. Each carries a mark as indicated.

Unless specifically mentioned—

Common data: Use the following common data (250nm technology) *if not mentioned specifically in the question*

Take $V_{DD} = 2.5\text{ V}$, $L_{min} = W_{min} = 250\text{ nm}$, $R_{nmos} = 13\text{ kohm/ unit (W/L)}$, $R_{pmos} = 31\text{ kohm/ unit (W/L)}$,

$V_{TON} = 0.4\text{ V}$, $V_{TOP} = -0.43\text{ V}$, $\lambda_n = 0.1\text{ V}^{-1}$, $\lambda_p = 0.1\text{ V}^{-1}$, $u_n C_{ox} = 120\text{ uA/V}^2$, $u_p C_{ox} = 40\text{ uA/V}^2$

Gate capacitance of minimum size MOSFET $C_{g,min} = 0.05\text{ pF}$

- Use **square law current equation** unless specifically mentioned. Use lambda-based design rules unless specially mentioned.
- Neglect body effect, channel length modulation if not mentioned specifically.
- The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

Answer all the subparts of a question together in sequence.

Q1. For a digital CMOS circuit, the connection of MOSFETs in n-diffusion region of a pull-down network is given in **Fig.1**. Further, this circuit was fabricated in **90 nm** technology node. Consider the **W/L=20** for each transistor used in this diagram. Now, Answer the following question:

Given: $C_{j0} = 2.8 \times 10^{-4}\text{ pF/}\mu\text{m}^2$, $C_{jsw} = 1.8 \times 10^{-4}\text{ pF/}\mu\text{m}$

- Find the length and width of each transistor in nm for the given layout.
- Label the source and drain on the layout corresponding to each transistor.
- Draw the circuit level schematic of this pull-down network and label the node C1 & C2 on it.
- Calculate the parasitic junction capacitance at the output node under the state of thermal equilibrium when no external bias is applied. (Ignore the sidewall capacitance in channel region).
- Calculate the parasitic junction capacitance contributed by node C1 & C2 individually under the state of thermal equilibrium when no external bias is applied. (Ignore the sidewall capacitance in channel region).

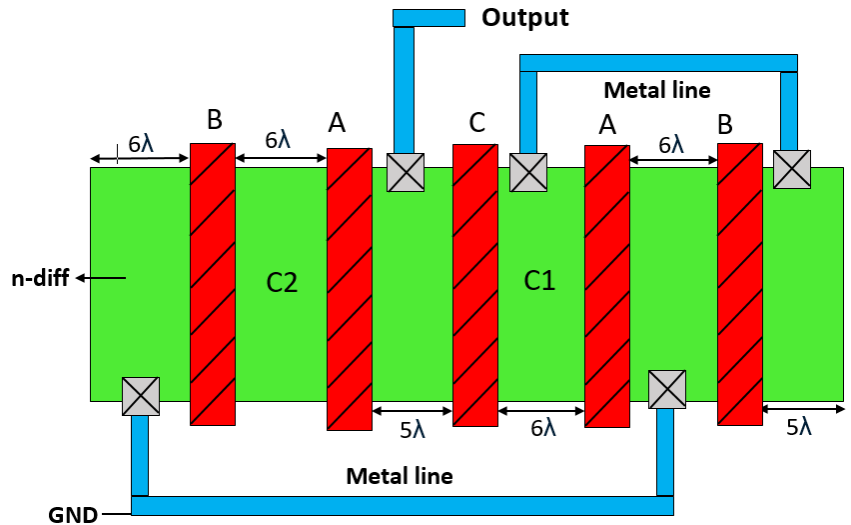


Fig. 1

[19 marks]

Q2. Given a CMOS inverter whose VTC (piecewise) shown in **Fig. 2**.

Use the following parameter values to design this inverter: $V_{tn} = |V_{tp}| = 0.4V$, $V_{DD} = 1.6V$, $\mu_n C_{ox} = 3\mu_p C_{ox} = 280\mu A/V^2$.

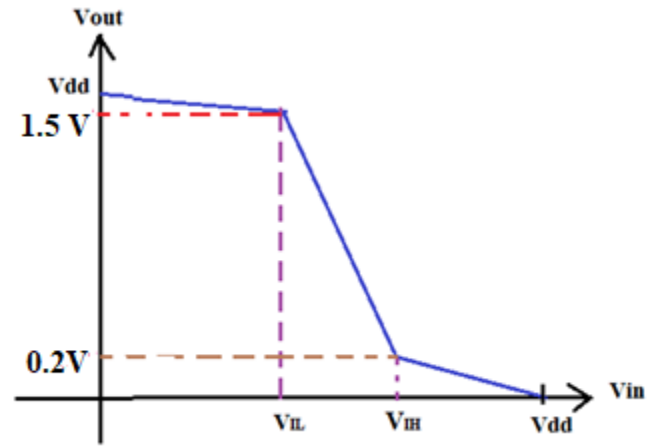


Fig. 2

- Suppose this inverter is going to drive a load capacitance of 24fF with a constraint of propagation delay $t_{PLH} = t_{PHL} \leq 22ps$. Determine the sizes (W/L) of the transistors to meet the above constraint. Use average current method.
- Also, calculate noise margins (low and high) of this inverter circuit.
- Determine the Energy drawn from the power supply per cycle. Also, Find out the dynamic power consumption of this inverter.

[19 marks]

Q3. A) Circuit shown in **Fig. 3 (a)** and (b) describes the 2-input CMOS NAND and NOR gates respectively, have been designed with aspect ratios of $(W/L)_n = 4$ and $(W/L)_p = 8$,

Given process variables $\mu_n C_{ox} = 140 \mu A/V^2$, $\mu_p C_{ox} = 90 \mu A/V^2$, threshold voltages $V_{tn} = 0.5 V$, $V_{tp} = -0.8 V$, supply voltage $V_{DD} = 3V$ with $C_{out} = 8fF$, and $C_x = 2 fF$.

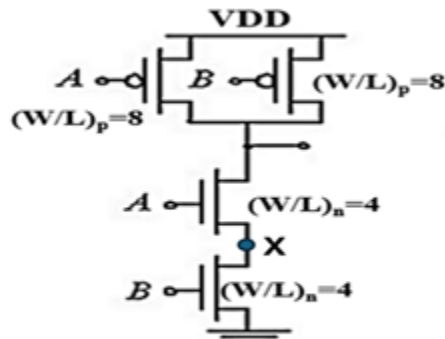


Fig. 3(a)

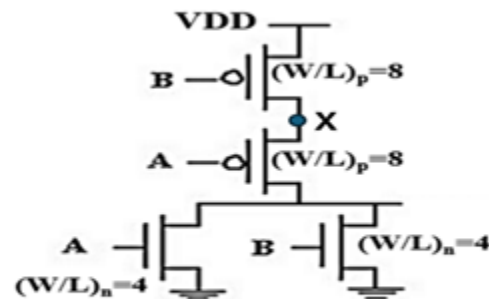


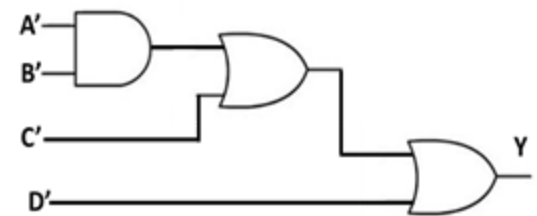
Fig. 3(b)

- Calculate R_p and R_n of PMOS and NMOS, respectively. Assume all the transistors are in linear region throughout the cycle.
- Calculate the worst-case and best-case rise time (10-90%) and fall time (90-10%) for a 2-input CMOS NAND gate. Use the values calculated in part a).

c) Calculate the switching activity factor ($\alpha_{0 \rightarrow 1}$) of a 2-input CMOS NOR gate. Then find the dynamic power consumption of a 2-input CMOS NOR gate that drives an **8 fF** output capacitance at **100 MHz**.

d) If the CMOS gate-based integrated circuit (IC) consumes **130 mW** total power at a **15 MHz** clock frequency. The same IC operating at a **10 MHz** clock frequency consumes **100 mW** total power. Determine the leakage power consumption of the IC neglecting short circuit power consumption.
Note—Here assume average switching activity factor and load per gate is α_{av} , and C_{av} respectively.

Q3. (B) Consider the gate-level diagram of a circuit in Fig. 4. Find the Boolean expression of the output signal Y. Draw the one stage transistor level CMOS based circuit diagram of signal Y for minimum parasitic capacitance at the output.



[19 marks]

Fig. 4

Q4. a) Consider the short and the long channel MOSFETs shown in Fig. 5. Let the width of the zero bias depletion regions is **0.35 μm** for the channel lengths of **1 μm** (short channel device) and **100 μm** (long channel device).

a) Determine the maximum width of the drain depletion widths so that the device operates normally. Assume a n-channel silicon MOSFET with $N_D \gg N_A$ and the source and substrate contacts are grounded.

b) For the short channel MOS devices shown in Fig. 5, derive the expression for the trans-conductance.

c) For the long channel MOS devices shown in Fig. 5, If both the lateral and vertical dimensions of the MOSFET are reduced by 30%, and the supply voltage (V_{DD}) and threshold voltage (V_{th}) are also reduced by 30%, calculate the percentage reduction/increases in the dynamic power dissipation. Assume constant electric field scaling.

d) Repeat part (c) if the gate oxide thickness is not scaled.

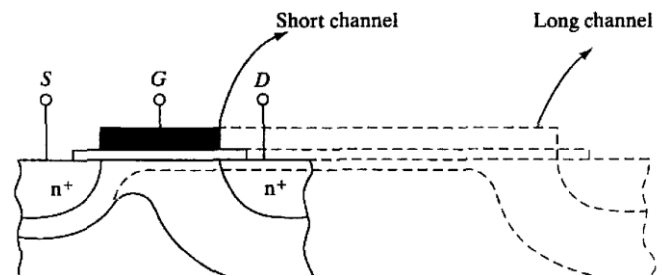


Fig. 2

[18 marks]

-----end-----